arXiv:2410.09584v1 [cs.CL] 12 Oct 2024

Toward General Instruction-Following Alignment for Retrieval-Augmented Generation

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Abstract

Following natural instructions is crucial for the effective application of Retrieval-Augmented Generation (RAG) systems. Despite recent advancements in Large Language Models (LLMs), research on assessing and improving instruction-following (IF) alignment within the RAG domain remains limited. To address this issue, we propose VIF-RAG, the first automated, scalable, and verifiable synthetic pipeline for instruction-following alignment in RAG systems. We start by manually crafting a minimal set of atomic instructions (<100) and developing combination rules to synthesize and verify complex instructions for a seed set. We then use supervised models for instruction rewriting while simultaneously generating code to automate the verification of instruction quality via a Python executor. Finally, we integrate these instructions with extensive RAG and general data samples, scaling up to a high-quality VIF-RAG-QA dataset (>100k) through automated processes. To further bridge the gap in instruction-following auto-evaluation for RAG systems, we introduce FollowRAG Benchmark, which includes approximately 3K test samples, covering 22 categories of general instruction constraints and four knowledge-intensive QA datasets. Due to its robust pipeline design, FollowRAG can seamlessly integrate with different RAG benchmarks. Using FollowRAG and eight widely-used IF and foundational abilities benchmarks for LLMs, we demonstrate that VIF-RAG markedly enhances LLM performance across a broad range of general instruction constraints while effectively leveraging its capabilities in RAG scenarios. Further analysis offers practical insights for achieving IF alignment in RAG systems. Our code and datasets are released at https://FollowRAG.github.io.

1. Introduction

The advancement of Large Language Models (LLMs) (OpenAI 2023; Yang et al. 2024) has profoundly revolutionized a variety of real-world tasks expressed in natural language (Wei et al. 2022; Luo et al. 2023). However, they still suffer from hallucinations and factual inconsistencies (Bang et al. 2023), impacting the authenticity of generated answers. Retrieval-Augmented Generation (RAG) has gained recognition as a promising solution, empowering LLMs to leverage reliable information from retrieved documents, thereby returning high-quality responses (Guu et al. 2020; Lewis et al. 2020).

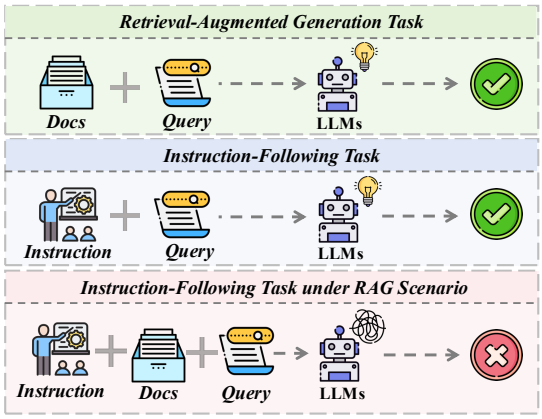

Figure 1: The task format of instruction-following tasks for LLMs in RAG scenarios.

In real-world interaction scenarios, users often deviate from standard templates when posing questions, instead of imposing diverse instructions on model outputs to meet specific task requirements (Jiang et al. 2023b; Chung et al. 2024). Consequently, improving instruction-following (IF) capabilities is foundational to the effective application of LLM and RAG systems. The core goal of IF is to enable models to adapt to the diverse intents of users, which has garnered widespread attention in the LLM community.

Existing efforts on instruction-following alignment primarily focus on multi-grained evaluation (Zhou et al. 2023a; Jiang et al. 2024a; Wen et al. 2024) and high-quality instruction data synthesis (Sun et al. 2024a; Zhao et al. 2024) to enhance LLMs' natural instruction-following capabilities. However, in complex RAG scenarios, the diverse knowledge introduced by retrieval-augmented techniques presents significant challenges for LLMs in effectively handling complex instructions (Figure 1). As shown in Figure 2, after supervised fine-tuning on high-quality general and knowledge-intensive QA datasets, LLMs demonstrate robust performance in both IF and RAG tasks (Mistral-base vs. Mistral-SFT). However, these capabilities do not always generalize well to instruction-following tasks under RAG scenarios and may even conflict with the performance of other fundamen-

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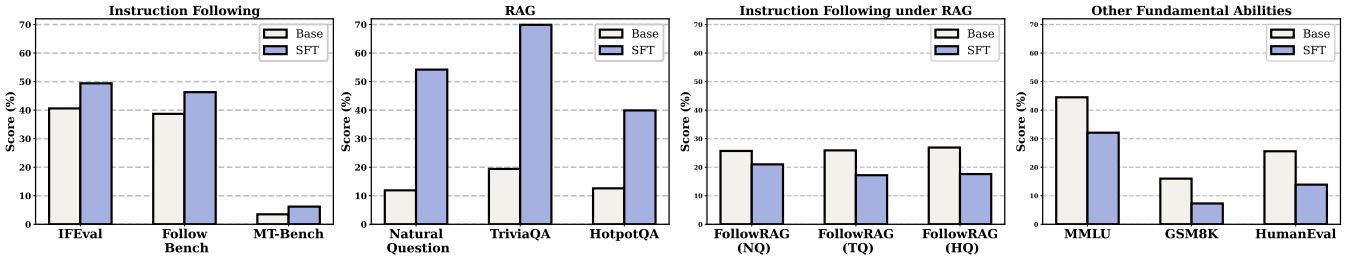

Figure 2: The performance comparison between Mistral-7B base and SFT version models on different tasks. The SFT version refers to the base model that has been fine-tuned using a mixed dataset, including NQ, TQ, HQ, and ShareGPT. Details results and setups can be found in Table 1 & 2

tal abilities (Dong et al. 2024b; Zhu et al. 2024). Unfortunately, research on instruction-following in RAG systems remains limited, significantly hindering their application in real-world interactions.

To tackle these challenges, our aim is to address following critical research questions:

- ***RQ1.*** *How can we comprehensively evaluate the complex instruction-following capabilities in the RAG scenario?*
- ***RQ2.*** *How can we achieve scalable and reliable instruction-following alignment in RAG systems while preserving the it's foundational abilities from conflict?*

In this paper, we propose VIF-RAG, the first automated, scalable, and reliable data synthesis pipeline for achieving complex instruction-following alignment in RAG scenarios. The core insight of VIF-RAG is to ensure every step of data augmentation and combination includes a proper verification process. Specifically, we start by manually crafting a minimal set of atomic instructions (<100) and developing combination rules to synthesize and verify complex instructions for a seed set. We then use supervised models for instruction rewriting. Motivated by tool execution studies (Le et al. 2022; Qiao et al. 2024b), we employ the same supervised model to generate verification code and automatically verify the quality of augmented instructions through the Python compiler's outputs. Finally, we combine these high-quality instructions with RAG datasets from various domains (each containing retrieved documents per query), performing the augmentation and dual validation process to synthesize a high-quality instruction-based RAG dataset, named VIF-RAG-QA (>100K samples).

To further bridge the gap in automatic instruction-following evaluation for RAG systems, we introduce FollowRAG, the first benchmark dedicated to comprehensively assessing the complex instruction-following capabilities of RAG systems. FollowRAG aggregates constraints from real-world scenarios. It includes approximately 3K test samples, spanning 4 knowledge-intensive QA benchmarks and 22 types of constraints. Due to its robust pipeline design, FollowRAG can seamlessly integrate with different RAG benchmarks.

To summarize, our contributions are as follows:

- To first achieve instruction-following alignment in the RAG system, we propose VIF-RAG, the first automated, scalable, and verifiable data synthetic framework. VIF-RAG uniquely combines augmented rewriting with diverse validation processes to synthesize high-quality instruction-following alignment data from almost scratch (<100), scaling up to over 100K samples.
- We introduce FollowRAG, the first benchmark designed to comprehensively evaluate LLM's complex instruction-following abilities in RAG tasks. FollowRAG includes nearly 3K test samples, spanning four knowledge-intensive QA benchmarks and 22 types of constraints. Its design ensures seamless integration with various RAG benchmarks, providing strong scalability.
- With FollowRAG and 8 widely-used IF and 3 foundational abilities benchmarks, we demonstrate that different LLMs with VIF-RAG achieve extraordinary alignment on general instruction following in both RAG and standard scenarios while effectively preserving other foundational capabilities. Further analysis offers practical insights for optimizing IF alignment in RAG systems.

2. Related Work

Instruction-Following Alignment for LLMs. Instruction-following ability is a core capability of large language models. Existing works fall into two main categories. The first includes efforts like MMLU and MTbench (Hendrycks et al. 2021; Zheng et al. 2024a), which rigorously evaluate models' adherence to general instructions. Moreover, works like IFEval and Followbench (Zhou et al. 2023a; Jiang et al. 2024a) focus on fine-grained assessment under specific constraints, using stricter criteria such as instruction difficulty, domain, and task formats (Qin et al. 2024; Xia et al. 2024; Yan, Luo, and Zhang 2024; Wen et al. 2024). The other category focuses on improving IF alignment. Manual design of instructions and responses by human annotators (Wei et al. 2021) is challenging and costly. To address this, methods are developed to synthesize diverse instructions, allowing weaker models to mimic the responses of advanced models (Dubois et al. 2024; Dong et al. 2024a; Xu et al. 2023), achieving strong-to-weak alignment (Cao et al. 2024).

Alignment for Retrieval-Augmented Generation. Retrieval-Augmented Generation (RAG) addresses the issue of knowledge hallucination in LLMs by retrieving relevant factual information, offering a promising solution (Guu et al. 2020; Lewis et al. 2020). However, efficiently aligning retrieved knowledge with LLMs' preferences remains a challenge. Researchers have developed robust reranker-based methods (Sun et al. 2023; Qin et al.

2023; Ma et al. 2023b) and data filtering approaches (Wang et al. 2023) to reduce noisy information and bridge this gap. Additionally, approaches like RePLUG (Shi et al. 2023) integrate LLMs' preferences into training objectives to improve alignment. Query rewriting methods (Ma et al. 2023a; Ren et al. 2023) attempt to adjust inputs based on these preferences. Furthermore, SelfRAG and MetaRAG (Asai et al. 2024; Zhou et al. 2024) use multi-round retrieval and generation to refine outputs and achieve better alignment. Despite these advancements, the diverse knowledge introduced by retrieval-augmented techniques poses significant challenges for LLMs in handling complex instructions. This highlights the need for further exploration into achieving effective instruction-following alignment in RAG systems.

3. Preliminaries

Retrieval-Augmented Generation (RAG). Retrieval-Augmented Generation systems usually operates under a *retrieve-then-read* framework (Lewis et al. 2020). The external retriever is integrated to gather supporting knowledge and improve the generation process. Given a query q, a retriever R recalls k relevant documents $D_q = \{d_i\}_{i=1}^{k}$ from an external corpus comprised of N documents. We employ the DPR (Karpukhin et al. 2020) to obtain hidden vectors for queries and documents. The relevance score is determined by measuring the dot-product similarity between the query and document representations, allowing the retrieval of the top-k documents D_q::

$$D_q = \text{argtop-}k \left[E_{\text{d}}(d_i)^{\top} \cdot E_{\text{q}}(q) \mid i = \{1 \ldots N\} \right]. \quad (1)$$

Then, the retrieved documents are concatenated with the query into an LLM reader R to generate the target text:

$$y = R(q, D_q) = \log P_{\theta}\left(q, D_q\right), \quad (2)$$

where P_{θ} is the output probability distribution.

Instruction-following Alignment for RAG. Following instructions is one of the most foundational ability for LLMs in RAG systems. Given an instruction $I = \{I_j\}_{j=1}^{M}$ with M specific constraints and a specific query q with corresponding relevant k retrieved documents D_q, The LLM π_{θ} in the RAG system is expected to produce an accurate response $y \sim \pi_{\theta}(y \mid q, D_q, I)$ while obeying with the specified constraints.

4. VIF-RAG Framework

In this section, we propose VIF-RAG, a verifiable automated instruction data synthesis framework for RAG scenarios. The core design of VIF-RAG is that each step of the automated generation or combination is accompanied by an appropriate verification process. ViF-RAG framework can be broadly split into two sections: (1) the instruction synthesis stage and (2) instruction-query synthesis, scaling from almost scratch (<100) to over 100K high-quality instruction-query samples. Below, we will delve into the specifics.

4.1. Instruction Synthesis from Scratch

Handwritten Seed Instructions. We initially develop a minimal seed instruction set $D_{\text{seed}}^{\text{atom}}$ manually, using four foundational categories of constraints: *format constraints, semantic constraints, knowledge constraints, and lexical constraints*, as themes for instruction writing. The following presents specific criteria regarding the 4 constraints:

- **Format Constraints** require the output to adhere to specific standards in terms of format, length, and structure. The content should be organized, clear, and meet the required format specifications.
- **Semantic Constraints** require the output's theme, language style, personality, and sentiment to align with the given instructions. The content should be semantically consistent with expectations and adhere to the specified tone or expression.
- **Knowledge Constraints** require the output to be accurate, comprehensive, and in-depth. The content should be informative, cover all necessary information, and maintain consistency in knowledge expression.
- **Lexical Constraints** require the output to include specific keywords or phrases, ensuring precision and relevance in word choice. The content should meet the expected requirements in terms of vocabulary selection.

We hire only one well-educated human annotator to manually create 15 single-atomic instructions for each type of constraint. Notably, this is the only process in our data synthesis process that includes human supervision.

Instruction Composition & Verification. Real-world instructions often involve multiple constraints in one user query. To address this complexity, we design rules to automatically combine atomic instructions into diverse, complex instructions:

- ***Multiple Constraints***: As illustrated in Figure 3, we randomly sample pairs of instructions from $D_{\text{seed}}^{\text{atom}}$ and insert them into a constraint template. By directly concatenating these instruction pairs, we create complex instructions that contain dual and triple constraints. This type of instruction requires the model to generate results that satisfy multiple constraints simultaneously.
- ***Chain Rule Constraints***: We design sequential conditional constraint templates and selected atomic instructions from $D_{\text{seed}}^{\text{atom}}$ to form chain constraints. Formally, the chain consists of n tasks $\{T_1, T_2, ..., T_n\}$, requiring the model's output to complete these n tasks sequentially.

Verification. Randomly combining these atomic instructions can easily lead to conflicts between them (e.g., don't use words containing the letter 'I', use words that end with '-ing'). These semantic conflicts can be challenging to detect using a simple Natural Language Inference model. To detect potential conflicts between these instructions, we use a robust supervised model that rates their consistency from 1 to 10. Samples scoring below 8 are excluded to refine our high-quality complex instruction set $D_{\text{seed}}^{\text{complex}}$. Ultimately, we arrive at the initial seed instruction set $D_{\text{seed}} = \{D_{\text{seed}}^{\text{atom}} \cup D_{\text{seed}}^{\text{complex}}\}$. Detailed information about the prompt templates are listed in the Appendix.

Instruction Rewriting & Quality Verification. To automate the scaling up of instructions, the instruction rewriting strategy is considered the most natural augmentation

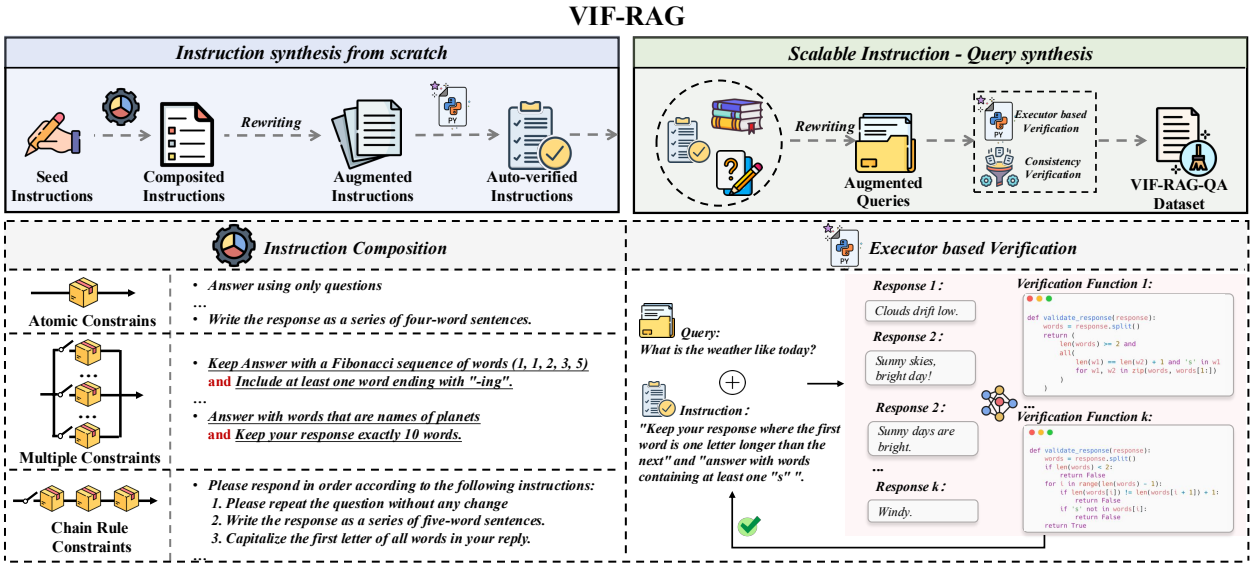

Figure 3: The overall framework of VIF-RAG. The top section presents the pipeline for automated IF data synthesis in RAG scenario, while the bottom section shows examples of 'Instruction Composition' and 'Executor-based Verification,' respectively.

method, and has received significant attention in the RAG and reasoning fields (Mumuni and Mumuni 2022; Xie et al. 2020; Yuan et al. 2023a; Li et al. 2024b,a). We use a supervised model[1] to iteratively rewrite instructions from the D_{seed} set in batches of 50 for K rounds, generating an augmented set D_{aug}. Subsequently, we merge the seed and augmented samples to form the combined instruction set $D_{\text{ins}} = D_{\text{seed}} \cup D_{\text{aug}}$, removing duplicates.

Inspired by tool execution works (Le et al. 2022), we aim to leverage the powerful coding abilities of LLMs to assist in verifying the quality of auto-generated instructions. As shown in Figure 3, for each instruction $I \in D_{\text{ins}}$, we use the supervision model to generate K verification function codes and corresponding test cases $\{func_j^I, c_j^I\}_{j=1}^K \in D_{\text{verify}}$, and assess the instruction's quality by analyzing the output of the executor $\mathcal{E}$. For any function and test case $\{func_j^I, c_j^I\} \in D_{\text{verify}}$, its execution output is:

$$\mathcal{E}(func_j^I, c_j^I) = \begin{cases} 1 & \text{If output is “True”} \\ 0 & \text{If output is “False” or “Error”} \end{cases} \tag{3}$$

Therefore, we can calculate the accuracy Acc_{func} of each verification function based on K test samples, as well as the accuracy Acc_{case} of each case evaluated using K verification functions. These can be formulated as:

$$\begin{cases} Acc_{\text{func}} = \frac{1}{K}\sum_{j=1}^{K} \mathcal{E}(func^I, c_j^I)_{j=1}^K \\ Acc_{\text{case}} = \frac{1}{K}\sum_{j=1}^{K} \mathcal{E}(func_j^I, c^I)_{j=1}^K \end{cases} \tag{4}$$

Based on the above cross metrics, we require that at least one Acc_{func} and Acc_{case} of the each instruction must exceed 0.5, Ultimately, we obtain the auto-verified instruction set as

$$D_{\text{ins}}^{\text{verify}} = \{d \in D_{\text{ins}} \mid Acc_{\text{func}}(d) > 0.5 \ \&\ Acc_{\text{case}}(d) > 0.5\} \tag{5}$$

[1]For the supervised model, we use GPT-4-turbo-2024-04-09. We conduct the ablation for supervision model in Table 7.

The samples that do not meet the cross metrics are discarded.

4.2. Scalable Instruction-Query Synthesis

Random Instruction-Query Combination. In real-world interactions with RAG systems, achieving IF alignment depends on effectively integrating the synthesized instructions with the queries used by the RAG system. To meet this goal, as depicted in Figure 3, we first extract high-quality queries from two different data sources.

1) RAG Domain: Building effective RAG system need to prepare sufficient amounts of QA-format data with relevant knowledge to enhance its knowledge-based interaction capabilities. Consequently, we randomly select a query set Q from mixed QA data sources, including open-domain multi-hop and knowledge base QA scenarios [2]. Following the *retrieve-then-read* paradigm (Lewis et al. 2020), We employ the dense retriever R to fetch the top-K relevant documents D_i for each query $q \in Q$ from an external knowledge base, resulting in the dataset $D_{\text{RAG}} = \{q_i, D_i\}_{i=1}^K$. Furthermore, we randomly select K queries along with their corresponding retrieved documents from D_{RAG} for each instruction I and combine them to create the RAG query set with IF constraints $D_{\text{IF-RAG}} = \{I_j, q_j, D_j\}_{j=1}^K$.

2) General Domain: In addition to incorporating RAG-specific abilities, the RAG system has to possess basic human-aligned abilities to meet users' daily interaction needs. Therefore, ShareGPT (Chiang et al. 2023), which provides authentic multi-turn human dialogue data, is our natural choice. Similar to how we handle the RAG domain, for each instruction $I \in D_{\text{ins}}$, we randomly select K queries from the ShareGPT to combine with the instruction and construct the general dataset $D_{\text{IF-General}}$ for each instruction.

Ultimately, we merge the instruction-constrained query

[2]We use the training sets from Natural Questions, TriviaQA, HotpotQA, and WebQuestionsSP as mixed QA sources.

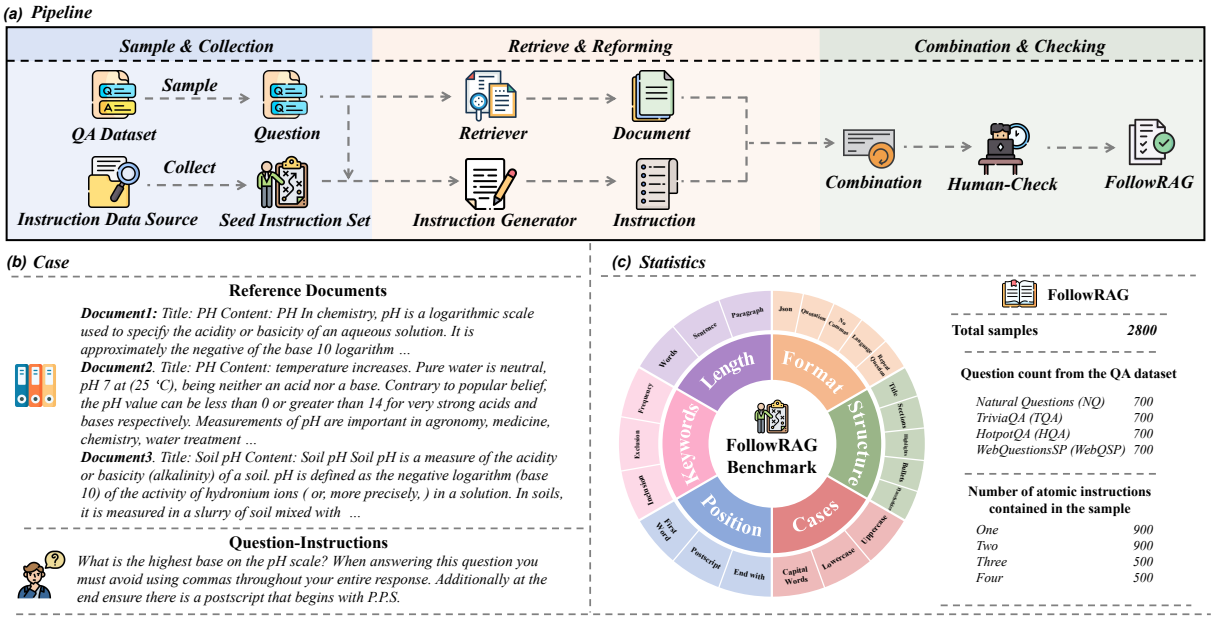

Figure 4: The construction pipeline, diagram and statistics of FollowRAG.

sets from these two domains into the final query set of VIF-RAG-QA, formulated as $D^q_{\text{VIF-RAG}}$.

Instruction-Query Rejection Sampling. It is worth noting that under diverse instruction-following constraints, the original grounding truth answers for queries in both the RAG and general datasets become unreliable. To address this issue and improve synthetic data diversity, we adopt a rejection sampling strategy (Yuan et al. 2023b). Specifically, we use the supervision model to generate K responses $y_x = \{y_i\}_{i=1}^K$ for each instruction-query pair $x \in D^q_{\text{VIF-RAG}}$, resulting in $\{x, y_x\} \in D_{\text{VIF-RAG}}$.

Dual Stage Verification. To further ensure comprehensive quality control of the synthetic dataset, we employ a dual stage verification process for the instruction-query data:

- ***Executor-based Verification:*** To automatically verify whether model-generated responses comply with the constraints of the instruction-query samples, we leverage pre-existing verification functions to evaluate adherence in the augmented outputs. As in the "Instruction Rewriting & Quality Verification" section, at least one response in $D_{\text{VIF-RAG}}$ must achieve an accuracy rate Acc_{case} above 0.5 across all verification functions; otherwise, the sample is discarded.
- ***Consistency Verification:*** We have noticed that combined instructions and queries often conflict. A simple example is when the query "Please write a brief biography of Barack Obama." does not meet the instruction "Strictly limit your answer to less than 10 tokens." Building on previous consistency verification of instructions, we employ a supervision model to evaluate the alignment between queries and instructions on a scale of 1 to 10, discarding samples that receive a score below 8.

After dual stage verification, we have automatically obtained a large-scale, high-quality VIF-RAG-QA dataset.

5. FollowRAG Benchmark

To bridge the gap in automatic instruction-following evaluation for RAG systems, we introduce FollowRAG in this section. We provide a detailed introduction from two aspects: "Data Construction" and "Evaluation and Statistics".

5.1. Dataset Construction

Instruction Collection & Extraction. FollowRAG aims to assess the model's ability to follow user instructions in complex multi-document contexts. Drawing from general IF datasets like IFEval (Zhou et al. 2023b) and FollowBench (Jiang et al. 2024b), we collect and verify definitions and examples of atomic instructions using rules (e.g., code), excluding those irrelevant to RAG scenarios. Ultimately, we identify 22 types of instruction constraints, encompassing language, length, structure, and keywords.

Instruction Reforming. We use widely-used question-answering (QA) datasets, such as Natural Questions (Kwiatkowski et al. 2019), as the foundation for constructing FollowRAG samples. For a query sampled from the QA datasets, we need to generate a complex instruction containing n atomic instruction constraints (with n ranging from 1 to 4). To enhance the diversity of atomic instruction representations, we employ GPT-4o as the instruction generator. Specifically, given a query, we first sample n instructions from the atomic instruction set and perform conflict detection. Subsequently, with examples as demonstrations, we prompt the LLM to generate a new varied instruction text for each type of atomic instruction, along with parameters for instruction-following evaluation.

Combination. Finally, we integrate the retrieved passages, query and atomic instructions to construct the sample input for FollowRAG. To avoid mechanically concatenating the query and instructions in a template-based manner, we prompt supervised model to naturally blend the multiple

atomic instructions and the query into a coherent instruction-query paragraph. We then add the top-K document passages retrieved based on the query to the instruction-query paragraph, forming the complete sample input.

5.2. Evaluation and Statistics

After obtaining the model's output, we evaluate it from two perspectives: instruction following and question answering (QA) under the RAG paradigm:

- ***Instruction Following:*** Utilizing the verifiable nature of our atomic instructions and following the IFEval approach, we automate the verification of the model's adherence to each instruction through code validation. We then calculate the average pass rate for each atomic instruction across all samples to determine the instruction-following score in FollowRAG.
- ***RAG:*** Under new instruction constraints, the model's target output differs from the gold answers in the original QA dataset, rendering traditional metrics like Exact-Match ineffective. To address this, we use the original gold answers as a reference and utilize GPT-4o to evaluate whether the model's outputs correctly address the questions. The scoring criteria are as follows: Completely correct (1 point), Partially correct (0.5 points), Completely incorrect (0 points). The average score of all samples is taken as the RAG score for FollowRAG.

For detailed statisticsin in Figure 4, FollowRAG is the first instruction-following evaluation dataset under RAG scenario comprising 2.8K samples, covering 22 fine-grained atomic instructions across 6 categories. The queries in FollowRAG are sourced from 4 QA datasets across 3 types: 1) Open-Domain QA: **Natural Questions (NQ)** (Kwiatkowski et al. 2019) and **TriviaQA (TQA)** (Joshi et al. 2017); 2) Multi-Hop QA: **HotpotQA (HQA)** (Yang et al. 2018); and 3) Knowledge Base QA: **WebQuestionsSP (WebQSP)** (Yih et al. 2016). To further construct varying levels of instruction-following difficulty, FollowRAG includes 0.9K samples of single and dual atomic instructions, as well as 0.5K complex multi-instruction samples containing 3 and 4 atomic instructions, respectively.

6. Experiment

6.1. Experimental Setup

Datasets. In this section, we evaluate over 10+ benchmarks to comprehensively evaluate the VIF-RAG. For the instruction-following tasks in RAG scenarios, we use the **FollowRAG** benchmark as mentioned in Section 5, which covering 4 question-answering (QA) datasets. For general instruction-following evaluation, we selected two commonly used complex instruction-following datasets, **IFEval** (Zhou et al. 2023a) and **FollowBench** (Jiang et al. 2024a), along with the natural instruction dataset **MT-Bench** (Zheng et al. 2024a) and the challenging ChatBot instruction-following bench, **Arena-Hard** (Li et al. 2024c). Additionally, to measure that the foundational abilities of LLMs, we further evaluate two widely used LLM's general abilties evaluation sets, **C-Eval** (Huang et al. 2023) and **MMLU** (Hendrycks et al. 2021), as well as the mathematical reasoning dataset **GSM8K** (Cobbe et al. 2021) and the code evaluation bench **HumanEval** (Chen et al. 2021).

For baselines, we select Mistral-7B (Jiang et al. 2023a), Llama3-8B (Meta 2024), Qwen1.5-7B, and Qwen1.5-14B (Yang et al. 2024) as our backbone models, fine-tuning ShareGPT and four QA training sets as SFT version. Besides, we introduce several strong IF baselines, including Conifer (Sun et al. 2024a), Evol-Instruct (Xu et al. 2023), and Deita (Liu et al. 2024). To ensure fairness, we add an equal-sized RAG training set to the original synthetic data used for these models. More details on the baselines and implementation can be found in the appendix.

6.2. Main Result

Our primary findings are presented in Table 1. Overall, VIF-RAG consistently surpasses all baselines in FollowRAG across multiple configurations, highlighting the clear advantages of our method. Additionally, we have discovered several key insights:

1) Existing IF baselines struggle in complex RAG scenarios. Comparisons between different base models and SFT versions in Tables 1 & 2 show that while SFT general data like ShareGPT improves performance on IFEval, it actually shows a performance decline in the instruction-following aspect of FollowRAG (e.g., NQ-IF: 25.7→21.0 in Mistral). Moreover, several strong IF baselines, such as Conifer (Sun et al. 2024b), also perform poorly in FollowRAG's IF aspect (HQ-IF: 26.9→26.45). This corroborates the issue highlighted in the introduction: traditional synthetic data may improve LLMs' vanilla instruction-following ability but often fails to generalize in RAG scenarios, sometimes even leading to decreased performance.

2) VIF-RAG shows exceptional IF alignment capability across various datasets, models, and parameter sizes. It consistently outperforms all baselines by over 10% on average accuracy, including a 44% improvement over Llama3-base, showcasing the significant performance advantage of our method. On four detailed QA benchmarks, VIF-RAG achieves the best results across all tested backbones. Moreover, whether using Qwen1.5-7B or Qwen1.5-14B, our method maintains a stable and significant performance increase of over 10%. These results highlight that VIF-RAG is not only plug-and-play but also exhibits strong generalization capabilities.

3) The RAG capability is effectively preserved. Protecting RAG capability is a core focus of RAG systems. Compared to various SFT version baselines, our VIF-RAG significantly enhances IF capability while maintaining more stable RAG performance. This allows us to be optimistic about its potential in real-world RAG system applications.

6.3. Cross-Domain Validation

To explore the transferability of VIF-RAG, we conduct cross-domain validation on four natural instruction-following datasets and four foundational abilities benchmarks for LLMs in Tabel 2. Our findings are as follows:

1) Consistent IF alignment in both standard and RAG scenarios. Table 1 shows that VIF-RAG achieves remark-

Model	NQ			TQ			HQ			WebQSP			ALL		
	IF	RAG	AVG	IF	RAG	AVG	IF	RAG	AVG	IF	RAG	AVG	IF	RAG	AVG
Llama3-8B-base	3.2	5.7	4.4	4.1	15.9	10.0	3.6	7.3	5.5	10.0	23.1	16.5	5.2	13.0	9.1
Llama3-8B-SFT	15.7	59.5	37.6	15.0	76.5	45.7	15.0	**52.5**	33.8	14.4	70.0	42.2	15.0	64.6	39.8
Llama3-8B-SFT-VIF-RAG	**43.9**	**65.0**	**54.5**	**42.7**	**78.0**	**60.4**	**39.6**	46.0	**42.8**	**42.5**	**70.5**	**56.5**	**42.2**	**64.9**	**53.5**
Mistral-7B-base	25.7	31.1	28.4	25.9	44.4	35.2	26.9	19.9	23.4	24.7	20.4	22.6	25.8	29.0	27.4
Mistral-7B-SFT	21.0	48.5	34.7	17.2	**71.5**	44.3	17.6	**46.5**	32.1	21.7	**66.5**	44.1	19.3	**58.3**	38.8
Mistral-7B-SFT Conifer	29.9	49.5	39.7	30.5	67.0	48.7	26.5	40.0	33.2	31.1	63.0	47.1	29.5	54.9	42.2
Mistral-7B-SFT Evol-Instruct	41.7	41.5	41.6	37.0	63.5	50.4	35.4	35.0	35.2	39.4	54.0	46.7	38.4	48.5	43.5
Mistral-7B-SFT-VIF-RAG	**51.2**	**56.5**	**53.8**	**45.9**	70.5	**58.2**	**44.9**	43.0	**44.0**	**47.8**	58.0	**52.9**	**47.4**	57.0	**52.2**
Deita-7B-V1.0-SFT	31.4	31.5	31.4	29.0	42.5	35.8	26.5	30.5	28.5	26.3	40.0	33.2	28.3	36.1	32.2
Qwen1.5-7B-base	27.7	34.4	31.0	27.7	45.9	36.8	27.5	19.8	23.6	29.9	45.8	37.9	28.2	36.5	32.3
Qwen1.5-7B-SFT	16.1	**50.5**	33.3	14.3	70.0	42.2	14.8	40.0	27.4	13.7	59.0	36.3	14.7	54.9	34.8
Qwen1.5-7B-SFT-VIF-RAG	**38.9**	41.5	**40.2**	**35.8**	**78.0**	**56.9**	**38.1**	**45.0**	**41.6**	**31.9**	**60.0**	**45.9**	**36.2**	**56.1**	**46.2**
Qwen1.5-14B-base	33.7	38.1	35.9	32.5	54.7	43.6	32.4	26.5	29.5	33.0	48.3	40.7	32.0	41.9	36.9
Qwen1.5-14B-SFT	22.0	**54.5**	38.3	18.7	66.0	42.3	18.8	**41.0**	29.9	19.9	63.0	41.4	19.8	56.1	38.0
Qwen1.5-14B-SFT-VIF-RAG	**42.1**	53.0	**47.6**	**40.1**	**71.0**	**55.5**	**38.8**	39.5	**39.2**	**35.7**	**69.0**	**52.3**	**39.2**	**58.1**	**48.6**

Table 1: The main results on FollowRAG."AVG" represents the weighted average of the corresponding IF and RAG scores. The top two results in each column are highlighted in **bold** and underlined.

Model	IFEval				FollowBench (SSR Avg.)	MT-Bench	Arena-Hard	C-Eval	MMLU	GSM8k	HumanEval (Pass@1)
	Pr (S)	Pr. (L)	Ins. (S)	Ins. (L)							
Llama3-8B-base	24.6	26.1	38.1	39.7	11.6	4.0	0.5	24.2	38.8	0.5	0.6
Llama3-8B-SFT	32.5	34.3	43.3	45.4	33.6	5.6	2.2	35.6	45.2	12.6	3.6
Llama3-8B-SFT-VIF-RAG	**37.0**	**42.7**	**48.8**	**54.2**	**49.2**	**6.2**	**3.2**	**39.6**	**49.6**	**22.9**	**8.0**
Mistral-7B-base	14.6	15.3	25.8	27.0	38.0	3.5	0.6	31.8	44.5	**16.0**	25.6
Mistral-7B-SFT	23.3	24.6	38.4	45.7	42.9	6.2	3.1	26.2	32.1	7.3	13.9
Mistral-7B-SFT-VIF-RAG	**34.6**	**41.0**	**46.3**	**52.0**	**53.4**	**6.5**	**3.6**	**33.0**	**49.6**	**16.0**	**32.9**
Qwen1.5-7B-base	25.1	27.9	37.8	40.6	38.7	5.4	3.2	72.8	58.3	50.6	36.0
Qwen1.5-7B-SFT	36.4	39.3	46.4	49.4	46.3	5.7	2.1	69.1	55.5	48.6	39.0
Qwen1.5-7B-SFT-VIF-RAG	**42.3**	**46.0**	**53.5**	**57.1**	**51.1**	**6.1**	**3.9**	**75.6**	**61.2**	**61.4**	**44.5**
Qwen1.5-14B-base	35.5	39.0	46.7	50.2	45.5	5.8	6.4	77.8	64.7	71.8	**59.1**
Qwen1.5-14B-SFT	38.4	41.7	49.4	52.6	49.8	6.0	6.5	76.2	62.0	71.5	58.5
Qwen1.5-14B-SFT-VIF-RAG	**46.3**	**49.9**	**60.0**	**62.2**	**56.3**	**7.3**	**7.0**	**79.5**	**66.5**	**73.8**	**59.1**

Table 2: The cross-domain validation on 4 general instruction-following (Left 4) and 4 foundational abilities (Right 4) benchmarks. Pr. and Ins. refer to the prompt level and instruction level metric, respectively. S or L denote the strict or loose metrics used in IFEval.

Model	FollowRAG (NQ)		IFEval	
	IF	RAG	Ins(L)	Prompt(L)
Mistral-7B-SFT-VIF-RAG	51.6	56.5	41.0	52.0
w/o Multiple Constraints	46.5 (-5.1)	52.3 (-4.2)	37.9 (-3.1)	48.6 (-3.4)
w/o Chain rule Constraints	49.2 (-2.4)	53.3 (-3.2)	39.2 (-1.8)	49.9 (-2.1)
w/o Executor based Verification	43.5 (-8.1)	56.1 (-0.4)	33.2 (-7.8)	47.6 (-4.4)
w/o Consistency Verification	47.6 (-4.0)	46.2 (-10.3)	38.4 (-2.6)	46.5 (-5.5)

Table 3: Ablation study on different designs of VIF-RAG.

able IF alignment in RAG scenarios. In Table 2, comparing Llama3-8B SFT version, VIF-RAG demonstrates strong gains on two widely-used IF benchmarks, IFEval and FollowBench, with improvements of 8.8% (Ins.L) and 15.5% respectively. It also maintains stable improvement across different parameter sizes (7B & 14B). These results confirm that VIF-RAG consistently enhances IF alignment in both RAG and standard scenarios.

2) Robust General IF Transferability. To assess general instruction-following alignment, we test VIF-RAG on challenging benchmarks Arena-Hard and MT-Bench. The results demonstrate that VIF-RAG maintains consistent alignment across various backbones, with a notable 1.3% improvement on MT-Bench for the 14B model. This reveals significant potential for larger models in achieving better natural instruction alignment.

3) Great Preservation of foundational Abilities. Previous research highlights that enhancing specific capabilities often compromises others (Dong et al. 2024b; Hui et al. 2024). As indicated in Table 2, VIF-RAG effectively preserves general capabilities (MMLU, C-Eval), math reasoning (GSM8K), and coding skills (HumanEval) across different configurations, with some slight performance improvements. This preservation is largely attributed to the integration of ShareGPT data in the synthesis process, demonstrating VIF-RAG's ability to balance diverse capabilities while maintaining broad applicability.

6.4. Quantitative Analysis

Ablation Study. To examine the effects of various components in VIF-RAG, we conduct an ablation study in Table

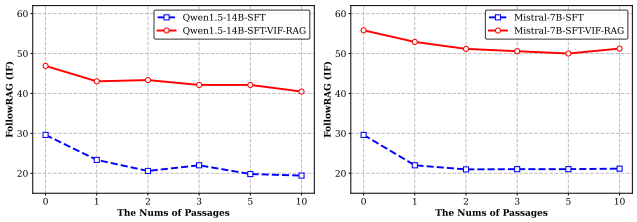

Figure 5: The scaling analysis of retrieved document count and FollowRAG (IF) performance.

3. The term "w/o" indicates versions where specific components are removed. Our key observations are:

- Removing any component from VIF-RAG results in decreased performance, indicating that all components, such as the complex instruction composition strategy and quality verification design, are crucial to its effectiveness.
- The largest performance decline in FollowRAG is observed when executor verification is removed. This underscores the critical role of automated instruction-response validation in improving synthetic data quality and confirms the advantage of using LLMs to oversee instruction-following abilities through other core skills like coding.
- Surprisingly, the consistency verification proves beneficial in preserving RAG capabilities. It effectively filters out samples with high-level semantic conflicts between instructions and queries, reducing noise in IF tasks and maintaining RAG performance integrity.

Scaling Analysis. To explore the impact of retrieved document quantity on instruction-following performance in RAG scenarios, we refer to Table 5. For the baseline models (SFT versions), instruction-following capability declines as the number of passages increases. Specifically, performance drops sharply by over 6% when the document quantity in FollowRAG increases from 0 to 1. Further increasing the number to 10 leads to a significant performance decline, with Qwen-14B-SFT experiencing a drop of over 10%. This indicates that integrating knowledge through retrieval-augmented techniques challenges the instruction-following abilities of existing models.

In contrast, VIF-RAG shows a minor performance drop (<3%) when encountering the first document. As the number of documents increases to 10, VIF-RAG's performance remains relatively stable, demonstrating its robustness.

Instruction Difficulty Analysis. To explore the effect of different instruction quantities (i.e., instruction-following difficulty) on model performance in RAG scenarios, we evaluate VIF-RAG and various baseline models on the FollowRAG benchmark, using test sets with 1, 2, and 3 instructions. As shown in Figure 6, as the number of instructions increases, all models generally show a decline in instruction-following capability, but VIF-RAG consistently outperforms the rest. Notably, even with 3 instructions present simultaneously, VIF-RAG still demonstrates over a 5% IF prompt (strict acc.), further validating its superior capability in handling complex instruction-following tasks in RAG scenarios.

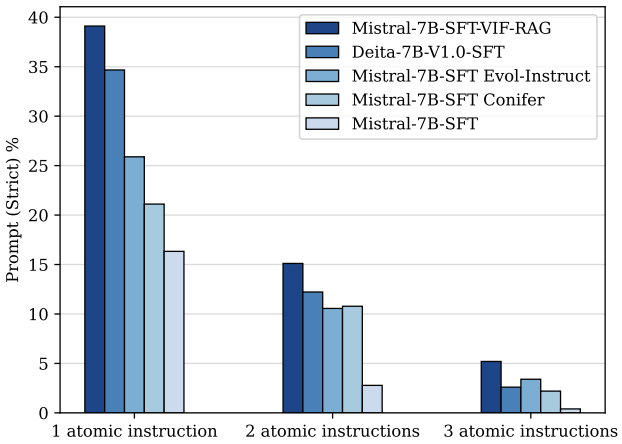

Figure 6: The analysis of instruction counts on FollowRAG (IF) performance.

7. Conclusion

In this paper, we propose VIF-RAG, the first automated, scalable, and verifiable data synthesis pipeline for aligning complex instruction-following in RAG scenarios. VIF-RAG integrates a verification process at each step of data augmentation and combination. We begin by manually creating a minimal set of atomic instructions (<100) and then apply steps including instruction composition, quality verification, instruction-query combination, and dual-stage verification to generate a large-scale, high-quality VIF-RAG-QA dataset (>100K). To address gaps in instruction-following evaluation for RAG systems, we present FollowRAG Bench, featuring around 3K samples with 22 types of complex instruction constraints. Using FollowRAG and 8 widely-used IF and foundational abilities benchmarks, we show that VIF-RAG significantly enhances alignment on general instruction constraints and effectively demonstrates the core abilities of LLMs. Further analysis offers insights for optimizing instruction-following alignment in RAG systems.

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Datasets and Baselines

Training Datasets

In the data synthesis process of VIF-RAG, we use the following open-source datasets:

- **Natural Questions (NQ)** (Kwiatkowski et al. 2019). Natural Questions is a comprehensive dataset created to train and evaluate automatic question-answering systems. It collects a large number of queries from Google's website and employs humans to annotate them using relevant knowledge from Wikipedia. The dataset contains 307,372 training examples, 7,830 for development, and 7,842 reserved for testing.
- **TriviaQA (TQA)** (Joshi et al. 2017). TriviaQA is an extensive and challenging text-based QA dataset comprising over 950,000 samples from 662,000 documents sourced from Wikipedia and other web pages. It is designed to test the limits of traditional QA systems, offering scenarios where answers aren't easily extracted through simple span prediction due to the lengthy and complex contexts.
- **HotpotQA (HQA)** (Yang et al. 2018). HotpotQA aims to enhance the system's ability to answer multi-hop questions and its robustness in integrating external knowledge. Unlike other QA datasets that often lack the complexity needed for training systems in reasoning and explanation, HotpotQA offers a new challenge with 113,000 question-answer pairs based on Wikipedia.
- **WebQuestionsSP (WebQSP)** (Yih et al. 2016). WebQuestionsSP is a resource developed to assess the impact of using semantic parse labels in knowledge base question answering. It extends the original WebQuestions dataset by adding SPARQL queries for 4,737 questions and includes "partial" annotations for 1,073 questions where a complete parse was either unachievable or the questions were poorly formed or needed descriptive answers.
- **ShareGPT** (Chiang et al. 2023). ShareGPT is a collection of around 90,000 conversations gathered via the ShareGPT API before it was discontinued. The dataset includes user prompts and responses from OpenAI's ChatGPT, providing valuable insights into human-AI interactions. It primarily features messages in English and other Western languages, showcasing the linguistic diversity of its users.

Evaluation Benchmarks

For cross-domain verification in this paper, we use the following public datasets for evaluation.

- **IFEval** (Zhou et al. 2023a). IFEval is the most commonly used comprehensive instruction-following evaluation set for LLMs. The dataset includes over 500 prompts aimed at testing how effectively LLMs perform specific, verifiable tasks. It covers 25 types of atomic instructions, each verifiable using simple, interpretable, and deterministic programs to determine if the responses adhere to the instructions.
- **FollowBench** (Jiang et al. 2023b). Followbench evaluates a model's ability to follow complex instructions by categorizing the instruction-following assessment into five different categories. It employs a multi-level mechanism to precisely enforce these constraints by evaluating the associated difficulty levels.
- **MT-Bench** (Zheng et al. 2024a). MT-Bench aims at evaluating multitask learning models, particularly in multi-turn dialogue and instruction-following tasks. It includes 80 high-quality multi-turn dialogue questions across eight common use cases: writing, role-playing, information extraction, reasoning, mathematics, coding, STEM knowledge, and humanities/social sciences. MT-Bench emphasizes challenging questions to effectively distinguish the capabilities of different models.
- **Arena-Hard** (Li et al. 2024c). Arena-Hard is a dataset used to assess the robustness of dialogue systems by testing their performance on challenging and diverse scenarios. It includes 500 carefully selected user queries that reflect complex real-world conversations, including language variations, spelling errors, and grammatical mistakes.
- **C-Eval** (Huang et al. 2023). C-Eval, as a comprehensive evaluation of general capabilities for Chinese large models, categorizes 13,948 test samples into 52 sub-domains and encompasses a difficulty system across four dimensions.. There is also a subset, C-Eval Hard, focusing on problems that demand advanced reasoning skills.
- **MMLU** (Hendrycks et al. 2021).

  MMLU, as the most widely used general capability evaluation set, covers assessments across 57 domains, including science and math, with a difficulty level spanning multiple tiers. It serves as a key tool for assessing language model performance across varied tasks.
- **GSM8K** (Cobbe et al. 2021). GSM8K is a classic mathematical reasoning evaluation dataset that primarily focuses on math problem-solving at the grade school level. It requires models to arrive at answers by outlining their reasoning paths. The dataset contains over 8,000 samples, with the training set comprising 7,473 samples and the test set containing 1,319 samples.
- **HumanEval** (Chen et al. 2021). HumanEval as a benchmark for evaluating code generation models, featuring 164 unique programming problems, each with about 9.6 test cases. It assesses the functional accuracy of generated code through these diverse test cases. HumanEval+ extends this by increasing the average number of test cases to 774.8 per problem, providing a more rigorous evaluation of code generation models.

Baselines

We compare the VIF-RAG framework with several strong baselines for the instruction-following task as follows:

- **Evol-Instruct** (Xu et al. 2023). Evol-Instruct is the publicly available WizardLM-Evol-Instruct dataset, which includes 143k samples consisting of a blend of Alpaca and ShareGPT evolved data. In accordance with the approach outlined in the original paper. To ensure a fair comparison, we combined their dataset with the same amount of RAG data (NQ, TQ, HQ, WebQ) as used in VIF-RAG.

- **Conifer** (Sun et al. 2024a). Conifer is an advanced language model designed to excel at following complex, constraint-based instructions. It stands out for its progressive learning approach, where tasks start simple and increase in complexity, allowing the model to handle intricate instructions effectively. The model's dataset was meticulously crafted using GPT-4 to ensure a diverse and challenging set of instructions. This makes Conifer particularly strong in real-world applications that require precise instruction adherence, setting it apart from other models in its ability to manage complex tasks. To ensure a fair comparison, we combined their dataset with the same amount of RAG data (NQ, TQ, HQ, WebQ) as used in VIF-RAG.

- **Deita-7B-V1.0-SFT** (Liu et al. 2023). Deita-7B is a data selection method that focuses on high-quality data selection and instruction fine-tuning. It utilizes the DEITA method, which combines complexity, quality, and diversity filtering to optimize model training. Despite using a smaller dataset, Deita-7B excels in various natural language processing tasks by effectively leveraging high-quality data.

- **SFT Version.** To build a strong baseline model and a fair comparison with VIF-RAG, we use the same amount of ShareGPT and RAG data (NQ, TQ, HQ, WebQ) as in VIF-RAG's data synthesis process, mixing them together to fine-tune (SFT) different baseline models. This resulted in the strong baseline models labeled as "Backbone-SFT" in the main experiments.

Additionally, We list all the backbone models mentioned in the articles here.

- **Llama3-8B** (Meta 2024). Llama3-8B is part of the open-source Llama3 series, developed by MetaAI, is the latest and most advanced model in the Llama series. It offers notable improvements over Llama 2, Especially supporting longer input windows. These upgrades enhance performance in contextual understanding and language generation, making Llama 3 a standout in the series.

- **Qwen1.5-7B & 14B** (Bai et al. 2023). Qwen1.5 is a significant release in Alibaba Cloud's Qwen series of large language models, featuring models with 7B and 14B parameters. It excels in multilingual tasks and supports long-context processing up to 32K tokens, making it ideal for applications like chatbots and language understanding. These enhancements made Qwen 1.5 a powerful tool for diverse AI applications.

- **Mistral-7B** (Jiang et al. 2023a). Mistral 7B, released in September 2023 by Mistral AI, is an efficient language model utilizing advanced techniques. Despite having only 7 billion parameters, it outperforms many LLMs with same param size, such as reasoning, mathematics, and code generation. The model is open-source, making it widely accessible and customizable for different applications.

Implementation Details

Details about Instruction-Query Synthesis

For the data synthesis part of RAG in section 4.2 "Scalable Instruction-Query Synthesis", following several RAG works (Oguz et al. 2022; Dong et al. 2023; Luo et al. 2024), we use DPR (Karpukhin et al. 2020) as retriever for encoding knowledges. We use it to retrieve the top-K (k=3) relevant documents from the Wikipedia (Vrandečić and Krötzsch 2014) retrieval corpus based on similarity.. In this paper, we randomly samples 60K ShareGPT samples and 40K RAG samples (10K each from NQ, TQ, HQ, and WebQ), then concatenate them with our high-quality synthetic instructions.

Finally, we refer to data templates from previous RAG studies (Wang et al. 2023; Ren et al. 2023; Dong et al. 2024c; Qiao et al. 2024a) and directly concatenate the instruction with the RAG dataset.

For the ShareGPT data, we directly concatenate the instruction with the query without using any specific templates. For consistency checks across multiple instructions in section "Instruction Composition & Verification", The prompt are listed here:

Prompt Template for Multi-Instructions Verification

You are an expert proficient in determining whether multiple instructions are suitable to be implemented as simultaneous constraints.

[Instructions]**{instruction}**

The text contains two or more instructions. Based on the semantic coherence and logical connection, assess whether these instructions are suitable to be implemented as simultaneous constraints. Please first conduct a thorough analysis and then assign a score ranging from 0 to 10 on the last line. A score of 0 indicates that the instructions are highly inappropriate to coexist, while a score of 10 signifies that the instructions are very suitable to serve as concurrent constraints. Please ensure that only a score is provided in the format Score: score without any additional content on the last line.

Our VIF-RAG's prompt templates, instruction data format, verfication code, test cases and more can be found in the supplementary materials.

Details about Supervised Fine-tuning

For all LLM fine-tuning, we use a global batch size of 128, an input window of 4096, and a learning rate of 7e-6 with 2% warm-up. Each set of experiments involves fine-tuning for 3 epochs. Our training framework is DeepSpeed Zero3 (Rasley et al. 2020). To reduce memory usage, we also employed the Flash Attention (Dao et al. 2022) strategy during training and utilized BF16 for mixed precision testing.

Our experiments are performed on NVIDIA A800 GPUs. Specifically, Qwen1.5-7B, Mistral-7B, and LLaMA3-8B are trained on 8 A800 GPUs. We use the Llama Factory framework (Zheng et al. 2024b) (version 0.6.3) for training and employed greedy decoding to test the HumanEval dataset, with the metric being Pass@1. We use five sets of random seeds to conduct the same series of experiments.

Knowledge Bases for RAG

For the NQ, TQ, and HQ datasets, We used Wikipedia as the retrieval knowledge base. We follow the DPR approach by first applying the pre-processing code from DrQA (Chen et al. 2017) to extract clean text, removing tables, infoboxes. Each article is then divided into 100-word text blocks, resulting in a total of 21,015,324 passages. Each passage is prefixed with the article title and an [SEP] token.

For WebQSP, we utilize Freebase (Bollacker et al. 2008) as the knowledge base, following the methods described in unikQA and SKP. Freebase, which includes over 125 million tuples, more than 4,000 types, and over 7,000 properties, is used for knowledge retrieval. To handle the challenge of indexing billions of relations, we implement a two-step retrieval process. DPR retrieves relations from this reduced set. The retrieved relations, typically short sentences, are combined into passages of no more than 100 tokens and provided to the FiD reader as text paragraphs.

Details of FollowRAG

Atomic instructions in FollowRAG

We present the 22 types of atomic instructions included in FollowRAG in Table 4.

Judging Prompt for RAG Scores in FollowRAG

Under multiple instruction constraints, the model's target output differs from the gold answers in the original QA dataset, rendering previous evaluation metrics like exact match ineffective. To address this issue, we use the original gold answers as the reference and employ GPT-4o to assess whether the model's output correctly answers the questions. The prompt used to instruct GPT-4o to evaluate the responses is as follows:

Judging Prompt for RAG Scores

Please act as an impartial judge and perform the task:
Given a [Question], you need to evaluate whether the [Response] correctly answers or hits the correct answer, and output your judgment after [Judge]. I will provide a correct answer [Reference] as a reference.
Scoring criteria:
- If the [Response] is completely correct and aligns with the correct answer, it scores 1 point;
- If the [Response] partially answers correctly, it scores 0.5 point;
- If the [response] is completely incorrect compared to the [Reference], it scores 0 point.

Note:
- Your only evaluation criterion is whether the [Response] correctly answered the answer, regardless of the format, language, case, length, etc., of the [Response]. Besides, providing more information than the [Reference] in the [Response] cannot be a reason for point deduction.
- Use the [Reference] as the correct answer reference rather than your own knowledge.
- The rating reply must strictly follow the format below: "Rating: [judge_score]\nReason: [judge_reason]", and do not output any other content. For example: "Rating: [0]\nReason: [Response and Reference are completely unrelated.]". Ensure that judge_score and judge_reason are enclosed in [].

[Question]
{question}

[Reference]
{answer_gold}

[Response]
{response}

[Judge]

Considering that evaluating the RAG scores for all samples in FollowRAG requires a substantial number of GPT-4o calls, we randomly sampled 100 entries from NQ, TQ, HQ, and WebQ for scoring and calculating the RAG scores.

Consistency with Human Evaluation

To evaluate the effectiveness of GPT-4 scoring in assessing LLM responses, we conducted a consistency experiment between GPT-4 prediction scores and human scores. For fill-in-the-blank and open-ended questions, we randomly sampled 30 instances each from the base model, the SFT version model, and the VIF-RAG model test cases, totaling 90 instances, and had a human annotator score these predictions. In Table 5, we report the consistency between the average human scores and GPT-4 scores, measured by Pearson correlation. The strong alignment between human and GPT-4 scores validates the effectiveness of GPT-4 scoring.

More Experiments for VIF-RAG

Details of Main Results

Since FollowRAG adopts the code-based instruction following verification method following IFEval, its instruction following metrics can correspond to the two levels in IFEval as well:

- **Instruction:** The proportion of followed atomic instructions to the total number of atomic instructions in the entire dataset.

Type	Name	Explanation
Keywords	Inclusion	Include specific keywords in the response.
	Exclusion	Exclude specific keywords in the response.
	Frequency	Frequency constraint for including specific keywords in the response.
Length	Words	Constraint on the number of words.
	Sentence	Constraint on the number of sentences.
	Paragraph	Constraint on the number of paragraphs.
Format	Json	Wrapped the response in JSON format.
	Quotation	Response wrapped in double quotes.
	No Commas	No commas allowed.
	Language	Restrict output language.
	Repeat Question	Repeat the question before answering.
Structure	Title	Include a specific title.
	Sections	Constrain the number of sections.
	Highlights	The answer must highlight at least {N} parts.
	Bullets	Constrain the number of bullet points.
	Placeholder	Constrain the number of placeholders.
Cases	Uppercase	Response must be in all capital letters.
	Lowercase	Response must be in all lowercase letters.
	Capital Words	Constrain the number of capitalized words
Position	End with	Response must end with specific content.
	Postscript	Use special markings at the end of the Response, such as P.S.
	First Word	Constrain the starting word of paragraph n.

Table 4: Names and explanations of the 22 types of atomic instructions included in FollowRAG.

Model	Qwen1.5-14B-base	Qwen1.5-14B-SFT	Qwen1.5-14B-SFT-VIF-RAG	ALL
Consistency	0.9639	0.9598	0.9619	0.9626

Table 5: The Pearson correlation coefficient between GPT-4o scoring and human scoring for the RAG score in FollowRAG.

Setup	Bench.	Train	Test	Rephrase	Percentage↓	N-gram↓
ShareGPT+RAG	FollowRAG	10K	2.8K	11	0.4%	5.3%
	IFEval	10K	542	2	0.05%	4.9%
	Followbench	10K	820	1	0.01%	2.7%
VIF-RAG-QA	FollowRAG	10K	2.8K	3	0.1%	3.1%
	IFEval	10K	542	0	0.01%	4.3%
	Followbench	10K	820	1	0.01%	2.6%

Table 6: Contamination analysis on VIF-RAG data. Train & Test denotes the size of corresponding set. Rephr. represents samples similar to the test sample

- **Prompt:** The proportion of samples where all atomic instructions are followed to the total number of samples in the entire dataset.

In addition, "Strict" and "Loose" indicate whether the response will be processed before scoring, such as removing common font modifiers, introductory phrases like "Sure, here it is:", and closing phrases like "Hope it helps." We adopt the Loose Instruction score in main text and present all the different instruction follwoing scores in Table 8.

Model	FollowRAG		IFEval	
	IF	RAG	Ins(L)	Prompt(L)
Qwen1.5-7B-base	28.2	36.5	27.9	40.6
Supervision Model: GPT-4				
Qwen1.5-7B-SFT-VIF-RAG	36.2	56.1	46.0	57.1
Supervision Model: Qwen2-72B				
Qwen1.5-7B-SFT-VIF-RAG	39.0	55.1	44.0	54.0
Supervision Model: Llama3-70B				
Qwen1.5-7B-SFT-VIF-RAG	40.6	52.3	41.0	52.3

Table 7: Ablation study on supervision models from GPT-4 with Qwen2-72B and Llama3-70B.

Data Contamination Analysis.

We evaluate the contamination of VIF-RAG-QA on FollowRAG, IFEval and FollowBench. Our detailed analysis is conducted separately from two aspects: rule-based detection and model-based detection.

For rule-based detection, we report contamination findings detected by traditional n-gram contamination algorithms. As shown in Table 6, both contamination rates are lower than those of the ShareGPT+RAG dataset we used.

For model-based detection, we employ LLM contamination detectors from LM-Sys (Yang et al. 2023), which utilize advanced chatbots to identify potentially rephrased contaminated test samples. Compared to ShareGPT+RAG dataset, Conifer shows relatively lower percentage of similar samples, which indicates an absence of data contamination. This

Model	NQ				TQ				HQ				WebQ				ALL			
	Pr. (S)	Pr. (L)	Ins. (S)	Ins. (L)	Pr. (S)	Pr. (L)	Ins. (S)	Ins. (L)	Pr. (S)	Pr. (L)	Ins. (S)	Ins. (L)	Pr. (S)	Pr. (L)	Ins. (S)	Ins. (L)	Pr. (S)	Pr. (L)	Ins. (S)	Ins. (L)
Llama3-8B-base	1.6	1.6	3.1	3.2	1.3	1.3	4.00	4.1	1.6	1.6	3.6	3.6	6.3	6.6	9.7	10.0	2.7	2.8	5.1	5.2
Llama3-8B-SFT	6.1	6.1	15.7	15.7	5.3	5.3	15.0	15.0	5.1	5.3	15.0	15.0	6.1	6.1	14.4	14.4	5.7	5.7	15.0	15.0
Llama3-8B-SFT-VIF-RAG	**22.3**	**23.0**	**43.1**	**43.9**	**20.0**	**20.1**	**41.7**	**42.7**	**18.6**	**19.3**	**38.3**	**39.6**	**19.0**	**19.3**	**41.6**	**42.5**	**20.0**	**20.4**	**41.2**	**42.1**
Mistral-7B-base	13.0	13.9	24.5	25.7	15.4	16.3	24.9	25.9	15.1	15.6	25.9	26.9	13.1	13.4	24.3	24.7	14.2	14.8	24.9	25.8
Deita-7B-V1.0-SFT	17.1	18.7	29.3	31.4	13.7	14.4	27.0	29.0	15.3	16.3	24.8	26.5	16.6	16.9	25.1	26.3	15.7	16.6	26.6	28.3
Mistral-7B-SFT Conifer	11.4	14.1	25.0	29.9	10.3	12.3	26.6	30.5	8.4	10.3	22.5	26.5	13.0	15.4	27.4	31.1	10.8	13.1	25.4	29.5
Mistral-7B-SFT Evol-Instruct	12.7	21.0	30.7	41.7	12.3	17.0	28.0	37.0	11.0	15.1	27.7	35.4	14.1	18.9	31.8	39.4	12.5	18.0	29.5	38.4
Mistral-7B-SFT	6.7	8.9	15.9	21.0	5.7	6.6	14.9	17.2	5.3	6.3	15.7	17.6	7.1	8.1	18.2	21.7	6.2	7.5	16.2	19.3
Mistral-7B-SFT-VIF-RAG	**19.4**	**31.3**	**37.6**	**51.2**	**17.0**	**25.6**	**34.4**	**45.9**	**17.0**	**23.7**	**33.9**	**44.9**	**20.6**	**27.9**	**37.9**	**47.8**	**18.5**	**27.1**	**35.9**	**47.4**
Qwen1.5-7B-base	13.7	13.9	26.9	27.7	13.6	14.0	26.6	27.7	13.6	14.4	26.6	27.5	**16.6**	**17.4**	28.3	30.0	14.4	14.9	27.1	28.2
Qwen1.5-7B-SFT	6.1	6.1	16.0	16.1	5.4	5.4	14.3	14.3	4.9	4.9	14.8	14.8	6.1	6.1	13.6	13.7	5.6	5.6	14.7	14.7
Qwen1.5-7B-SFT-VIF-RAG	**20.3**	**20.9**	**37.3**	**38.9**	**18.7**	**18.6**	**34.5**	**35.8**	**19.3**	**20.0**	**36.5**	**38.1**	15.1	15.9	**30.5**	**31.9**	**18.3**	**18.9**	**34.7**	**36.2**
Qwen1.5-14B-base	17.6	18.3	32.7	33.7	17.7	18.1	31.8	32.5	15.7	16.4	31.6	32.4	17.1	17.6	31.7	33.0	17.0	17.6	32.0	32.9
Qwen1.5-14B-SFT	9.6	9.7	21.7	22.0	7.4	7.4	18.3	18.7	7.4	7.6	18.7	18.8	9.0	9.0	19.5	19.5	8.3	8.4	19.6	19.8
Qwen1.5-14B-SFT-VIF-RAG	**23.6**	**24.9**	**40.7**	**42.1**	**22.3**	**22.7**	**38.5**	**40.1**	**22.9**	**23.6**	**37.3**	**38.8**	**17.9**	**18.4**	**34.8**	**35.7**	**21.7**	**22.4**	**37.8**	**39.2**

Table 8: Detailed scores of instruction following under different metrics for FollowRAG. "Pr." and "Ins." represent Prompt and Instruction levels, while "S" and "L" denote Strict and Loose.

allows us to confidently assert that there is no contamination between the self-generated training samples and the test sets.

Ablation for Supervision Model.

Table 3 presents the results of replacing the supervision model from GPT-4 with Qwen2-72B and Llama3-70B. We observe that in the VIF-RAG framework, the stronger supervision model (GPT-4) demonstrates more effective strong-to-weak distillation alignment. However, Qwen2-72B and Llama3-70B also maintain solid performance, with accuracy consistency in IFEval loose prompts exceeding 50%. This highlights the flexibility and robustness of our VIF-RAG framework, which can adapt well to different supervision models.

Case Presentation and Analysis

Our VIF-RAG synthetic instruction data, code, test cases, and more can be found in the supplementary materials.

The Case Study of VIF-RAG

To gain a deeper understanding of how VIF-RAG achieves instruction-following alignment in RAG scenarios, we conducted a case study and manual analysis, as shown in the figure 7, 8 and 9.

Since there is no truly "gold response" after following the instructions, we can only use the original gold response from the RAG dataset as a reference.

Challenges and Future Work

In this paper, we first explore instruction-following alignment in RAG scenarios and develope a high-quality RAG instruction-following data synthesis framework, VIF-RAG, along with a comprehensive benchmark, FollowRAG. However, during our research, we encounter several more challenging scenarios:

Increased Number of Instructions: As shown in Figure 1, our experiments revealed that existing models can handle up to 4 instructions in RAG scenarios. Even in such cases, VIF-RAG still manages to correctly answer several questions, while other baseline models lose accuracy entirely. Therefore, the challenge of increasing the number of instructions remains significant. Effectively addressing the multi-instruction problem in RAG scenarios continues to be a promising direction with major implications for complex RAG interactions.

More Complex Instruction Types: As the first benchmark for RAG scenarios, FollowRAG provides a comprehensive evaluation of existing models' accuracy in these scenarios. However, the variety of instructions in the real world is vast, and it is impossible to cover all types in one work. Further evaluating and improving the handling of complex instructions that are difficult to validate (e.g., interaction styles, domain-specific knowledge in RAG) will be an important focus for our future research.

We believe that future work on instruction-following alignment will offer greater promise for the practical applications of RAG systems.

Case 1

Instruction

Please ensure your response ends with a postscript starting with NB:. Your answer should consist of exactly 5 paragraphs, with paragraphs separated by two line breaks. Make sure that the first paragraph begins with the word "Abbey". Also, ensure that your answer includes at least 100 words. Finally, end with the phrase: Stay safe and take care.

Query

Where is Fulda and what is its significance?

Reference Documents

Document1: *Title: Fulda Content: Fulda Fulda () (historically in English called Fuld) is a city in Hesse, Germany; it is located on the river Fulda and is the ...*

Document2: *Title: Fulda monastery Content: Fulda monastery Fulda Abbey, or the Princely Abbey of Fulda, or the Imperial Abbey of Fulda (German: , ,) was a Benedictine abbey as well as an ecclesiastical principality centered on ...*

Document3: *Title: Zelda Fitzgerald Content: for the very first "Zelda" title." New York City's borough of Manhattan's Battery Park's resident ...*

Ref. RAG Answer

Hesse, Germany/on the river Fulda

Model's Outputs & Scores

Model	Response	RAG score	IF score
SFT Baseline	*Fulda is a town in Hesse, Germany. It is the administrative seat of the Fulda district.*	0.5	0.0
Evol-Instruct	*Abbey: Fulda Abbey, or the Princely Abbey of Fulda, orthe Imperial Abbey of Fulda (German: , ,) was a Benedictineabbey as well as an ecclesiastical principality centered on Fulda, inthe present-day German state of Hesse. It was founded in 744 by SaintSturm, a disciple of Saint Boniface. Through the 8th and 9thcenturies, Fulda Abbey became a prominent center of learningand culture in Germany, and a site of religious significance andpilgrimage following the burial of Boniface. The growth in populationaround Fulda would result in its elevation to a prince-bishopric in thelate 8th century. Significance: Fulda Abbey played a significant role in the reorganization of the church in Germany by Saint Boniface. It served as a base from which missionaries could accompany Charlemagne's armies in their politicaland military campaigns to fully conquer and convert pagan Saxony. The abbey also became a center of learning and culture, attracting scholars and intellectuals from all over Europe*	0.5	0.25
Conifer	*Abbey of Fulda*	0.5	0.0
Deita	*ABBEY*	0.0	0.0
VIF-RAG	***Abbey: Fulda Abbey, or the Princely Abbey of Fulda, or Imperial Abbey of Fulda, was a Benedictine abbey as well as an ecclesiastical principality centered in Fulda, Germany. Founded in 744 by Saint Sturm, a disciple of Saint Boniface, it became a prominent center of learning and culture. Location: Fulda is located in Hesse, Germany, on the river Fulda. It is the administrative seat of the Fulda district and hosted the 30th Hessentag state festival in 1990.*** ***Significance: Through the 8th and 9th centuries, Fulda Abbey was a site of religious significance and pilgrimage following the burial of Boniface. It served as a base for missionaries accompanying Charlemagne's armies in their campaigns to convert pagan Saxony.*** ***Germany: Fulda is part of Germany's rich history, with the abbey playing a significant role in the country's religious and cultural development.*** ***Conclusion: Stay safe and take care.***	**1.0**	**0.5**

Figure 7: A case study (1) of VIF-RAG and different models in the followrag benchmark, with the backbone Mistral-7B.

Case 2

Instruction

Please use only lowercase letters and ensure it is written in english. Ensure the entire output is enclosed within json format. using markdown ticks (`) is acceptable.

Query

What are dallas cowboys colors?

Reference Documents

Document1: *Title: Content: Dallas Cowboys sports team colors Blue . Dallas Cowboys sports team colors White . Dallas Cowboys sports team colors Royal blue . Dallas ...*
Document2: *Title: Content: Dallas Cowboys sports team roster from 1960 , and sports team roster player Don Meredith , and sports team roster position Quarterback ...*
Document3: *Title: Content: Dallas Cowboys sports team roster player Dallas Walker . Dallas Cowboys sports team roster /m/0j9_131 . Dallas Cowboys sports team roster to 1970 . Dallas Cowboys sports team ...*

Ref. RAG Answer

Royal blue/Silver/Blue/White/Navy Blue

Model's Outputs & Scores

Model	Response	RAG score	IF score
SFT Baseline	*Royal blue*	0.5	0.0
Evol-Instruct	*Royal blue*	0.5	0.0
Conifer	*navy blue*	0.5	0.5
Deita	*None*	0.0	0.0

VIF-RAG	```*json*
	{
	"dallas cowboys colors": "Blue, White, Royal blue, Navy Blue, Silver"
	}
	``
		1.0	**0.5**

Figure 8: A case study (2) of VIF-RAG and different models in the followrag benchmark, with the backbone Mistral-7B.

Case 3

Instruction

Make sure to surround your complete response with double quote marks. Additionally, ensure the word "Borgia" appears fewer than 4 times in your answer.

Query

Which infamous character became Pope Alexander VI?

Reference Documents

Document1:*Title: Cantarella (manga) Content: and is aided in his political machinations by the assassin Don Michelotto. Cantarella (manga) Cantarella is the ...*
Document2: *Title: Poisons or the World History of Poisoning Content: the influence of Prokhorov's stories about famous poisoners Oleg often "daydreams", imagining crimes of these villains ...*
Document3: *Title: Cantarella (manga) Content: Cantarella (manga) Cantarella is the story of Cesare Borgia, an Italian aristocrat during the Renaissance. In the manga, Cesare's father Rodrigo...*

Ref. RAG Answer

Rodrigo de Borja y Dams/Alexander VI Borgia/Rodrigo de Lanzol-Borgia/Cardinal de Borgia/Alexander Vi/Rodrigo Lanzol y Borgia/Rodrigo de Lanzòl-Borgia/Alexander Sextus/Alejandro VI, Catalan/Alexander vi/Rodrigo Borgia/214th pope/Rodrigo de Borja y Doms/Alexander VI/Pope Alexander VI/Rodrigo Lanzol y de Borja/Rodrigo Lanzol y Gorja/Roderick Borgia/Rodrigo de Borja y Borja/Roderic Llançol/Alexander The Sixth/Roderic de Borja i Borja/Roderic de Borja/Pope Alex VI/Rodrigo Lanzol-Borja y Borja/Rodrigo de Borja/Roderic Llançol i de Borja

Model's Outputs & Scores

Model	Response	RAG score	IF score
SFT Baseline	*Rodrigo Borgia*	1.0	0.5
Evol-Instruct	*"The Devil's Advocate" (film)*	0.0	0.5
Conifer	*Rodrigo de Borja y Austria-Lorena*	0.0	0.5
Deita	*Rodrigo*	0.5	0.5
VIF-RAG	***"Rodrigo Borgia"***	**1.0**	**1.0**

Figure 9: A case study (3) of VIF-RAG and different models in the followrag benchmark, with the backbone Mistral-7B.